Higher National Diploma in Engineering (Civil/Electrical/Mechanical)

First Year, Second Semester Examination - 2022

22EN1214 ENGINEERING MECHANICS

Question 1

a) The coordinates of a moving particle is given by $X = 4t - t^2/2$ and $Y = 1 + 3t - 5t^2$. Find the velocity and the acceleration of the particle when time $t = 2$ second. (10 marks)

b) The following forces act at a point:

- 50N inclined at 30° towards North of East
- 20N towards North
- 30N towards North West
- 35N inclined at 40° towards South of west

Find the magnitude and direction of the resultant force. (10 marks)

Question 2

a) Prove that the centroid of a quarter circle with radius r is located at $(\frac{4r}{3\pi}, \frac{4r}{3\pi})$. (10 marks)

b) A thin plate in a vertical plane has the shape of a quarter circle and is hinged at point O. The mass of the plate is m. A force F acting horizontally at point A maintains the plate's orientation as shown in the Fig. 2. Find the force F in terms of the mass m of the plate. (10 marks)

Question 3

a) Explain **stress, strain**, and the **modulus of elasticity**. Discuss their relationship according to **Hooke's Law**. (10 marks)

b) Three bars made of **copper, zinc**, and **aluminum** are rigidly connected at their ends and have equal lengths as shown in the Fig. 3. The cross-sectional areas of the bars are 600, 900 and $1200\ \text{mm}^2$ respectively. If the compound member is subjected to a longitudinal pull of $300\ \text{kN}$, estimate: i. The load carried by each rod. ii. The induced stresses in each material.

Given: Modulus of elasticity for copper $E_{Cu} = 1.3 \times 10^5\ \text{N/mm}^2$, for zinc $E_{Zn} = 1.0 \times 10^5\ \text{N/mm}^2$, and for aluminium $E_{Al} = 0.8 \times 10^5\ \text{N/mm}^2$. (10 marks)

Question 4

a) **Friction** can be both desirable and undesirable in engineering applications. Discuss its advantages and disadvantages with examples. (10 marks)

b) An inclined plane with an angle $\alpha = 20^\circ$ has a spring with a force constant $k = 150\ \text{N/m}$ fastened securely at the bottom, such that the spring is parallel to the surface. A block of mass $m = 1.5\ \text{kg}$ is placed on the plane at a distance $d = 0.5\ \text{m}$ from the spring. From this position, the block is projected downward toward the spring with a speed $v = 1\ \text{m/s}$ as shown in Fig. 4. By what distance is the spring compressed when the block momentarily comes to rest? (10 marks)

Question 5

A **screw jack** is used to raise a load in an engineering application. The screw jack has a threaded screw with a mean diameter of $40\ \mathrm{mm}$ and a pitch of $10\ \mathrm{mm}$. The coefficient of friction between the screw and the nut is 0.1. A load of $20\ \mathrm{kN}$ needs to be raised using this screw jack, with the assumption that the load rotates with the screw. i) Calculate the **torque** required on the screw to raise the load of $20\ \mathrm{kN}$. ii) Determine the ratio of the torque required to raise the load to the torque required to lower the load. iii) Calculate the **efficiency** of the machine. (20 marks)

Question 6

A simply supported beam **AB**, shown in Fig. 6, is supported at points **A** and **B**. The beam carries a **uniformly distributed load** of $2.5\ \mathrm{kN/m}$ over the length **AC** and a **point load** of $15\mathrm{kN}$ at point **D**. Neglect the weight of the beam. i) Find the support reactions at A and B. ii) Derive the expressions for the **shear force** and the **bending moment** for each segment of the beam. iii) Draw the shear force and bending moment diagrams. iv) Find the maximum bending moment. (20 marks)

Question 7

a) A **flat belt** connects pulley A and motor B as shown in Fig. 7(a). The coefficients of static friction and dynamic friction between both pulleys and the belt are 0.25 and 0.20 respectively. Given that the maximum allowable tension in the belt is $600\ \mathrm{N}$, calculate the maximum torque that the belt can apply to pulley A. (10 marks)

b) Consider a mechanical system, as shown in Fig. 7(b), where a block of mass m is connected to a pulley with moment of inertia I and radius R. A spring with stiffness k is attached to the pulley. i) Write down the differential equation of motion for the system. ii) Determine the expression for the **natural frequency of vibration** for the system. (10 marks)